The Extremal Principle — problem 3.23

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Take the points A_1, B_1, C_1, respectively on the sides AB, BC, CA of the triangle ABC. Show that if $|AA_1| \leq 1$, $|BB_1| \leq 1$, $|CC_1| \leq 1$, then the area of the triangle is $\leq 1/\sqrt{3}$.